

Strategic Research Proposal: A Dynamic Multi-Sensor GeoAI Framework for Real-Time Landslide Susceptibility Evolution and Cross-Basin Transferability in Nepal

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Abstract

Landslide susceptibility mapping (LSM) in Nepal is currently dominated by static models that fail to capture the temporal dynamics of environmental triggers. This research proposes a novel GeoAI framework integrating Sentinel-1 SAR and Sentinel-2 optical synergy with a hybrid rainfall fusion strategy. By leveraging the 2021–2026 temporal range and focusing on cross-basin transferability (Arun to Trishuli), this study aims to provide physically consistent, explainable insights (via SHAP) for national disaster risk management.

1 Introduction and Rationale

Nepal’s Mid-Hills face chronic landslide hazards, exacerbated by intensive road construction and climatic anomalies. Current mapping efforts by agencies like NDRRMA are hindered by ”static snapshots” which do not account for post-seismic landscape healing or monsoon-driven soil saturation levels. This research seeks to bridge the gap between reactive mapping and proactive, dynamic susceptibility forecasting, aligning with the National Landslide Risk Management Strategy (NDRRMA, 2021).

2 Literature Review & Research Gaps

2.1 Recent Advances (2024–2025)

Recent studies in the Hindu Kush Himalaya (HKH) have transitioned to ensemble GeoAI. *Bhattarai et al. (2024)* demonstrated the effectiveness of hybrid CNN-SVM models in capturing complex non-linearities in the Chure region. Furthermore, *Hussain et al. (2025)* and *Gnyawali et al. (2024)* have highlighted the critical importance of Explainable AI (XAI) using SHAP values to increase model transparency for policy-makers and engineers. The integration of Sentinel-1 SAR for soil moisture proxies has been identified as a transformative frontier, as shown in the cross-regional studies of *Yu et al. (2025)*.

2.2 Identified Research Gaps

1. **Dynamic Sensitivity:** There is a lack of operational models integrating monthly Sentinel-1 SAR backscatter as a proxy for soil moisture in the Nepal mid-hills (*Dahal et al., 2025*).
2. **Transferability:** Models often lack generalization capabilities when transferred across basins with different lithological fingerprints, a challenge noted in recent benchmark studies (*Frontiers in Earth Science, 2025*).
3. **Temporal Continuity:** No major study has continuously modeled the evolution of risk across the high-anomaly period of 2021–2026, leaving a significant gap in our understanding of post-2020 landscape dynamics.

3 Methodology

3.1 Hybrid Rainfall Fusion

To address data discontinuities in the Department of Hydrology and Meteorology (DHM) Nepal records, we implement a linear bias correction for GPM IMERG data, following the weighted fusion logic proposed by *NASA GPM Core Teams (2024)*:

$$P_{fused}(t) = w_D P_{DHM}(t) + w_G [\alpha \cdot P_{GPM}(t)] \quad (1)$$

3.2 GeoAI Architecture

We utilize an **Ensemble XGBoost** model, optimized via Bayesian optimization (Optuna). The research includes a three-stage **Ablation Study** to quantify the information gain from SAR synergy. Interpretability will be provided using *SHAP* (Lundberg and Lee, 2017) to ensure geophysical consistency.

4 Bibliography

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